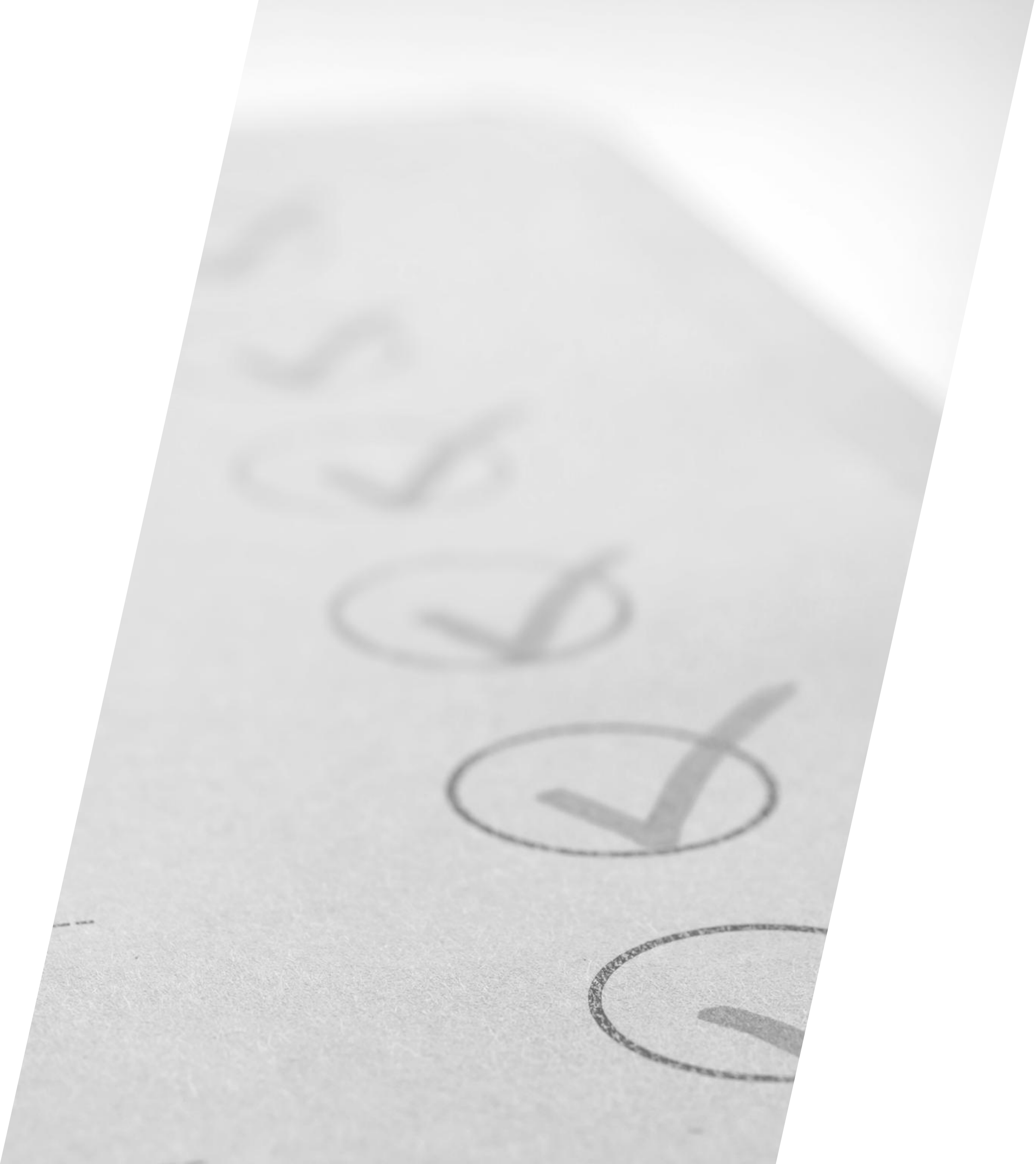
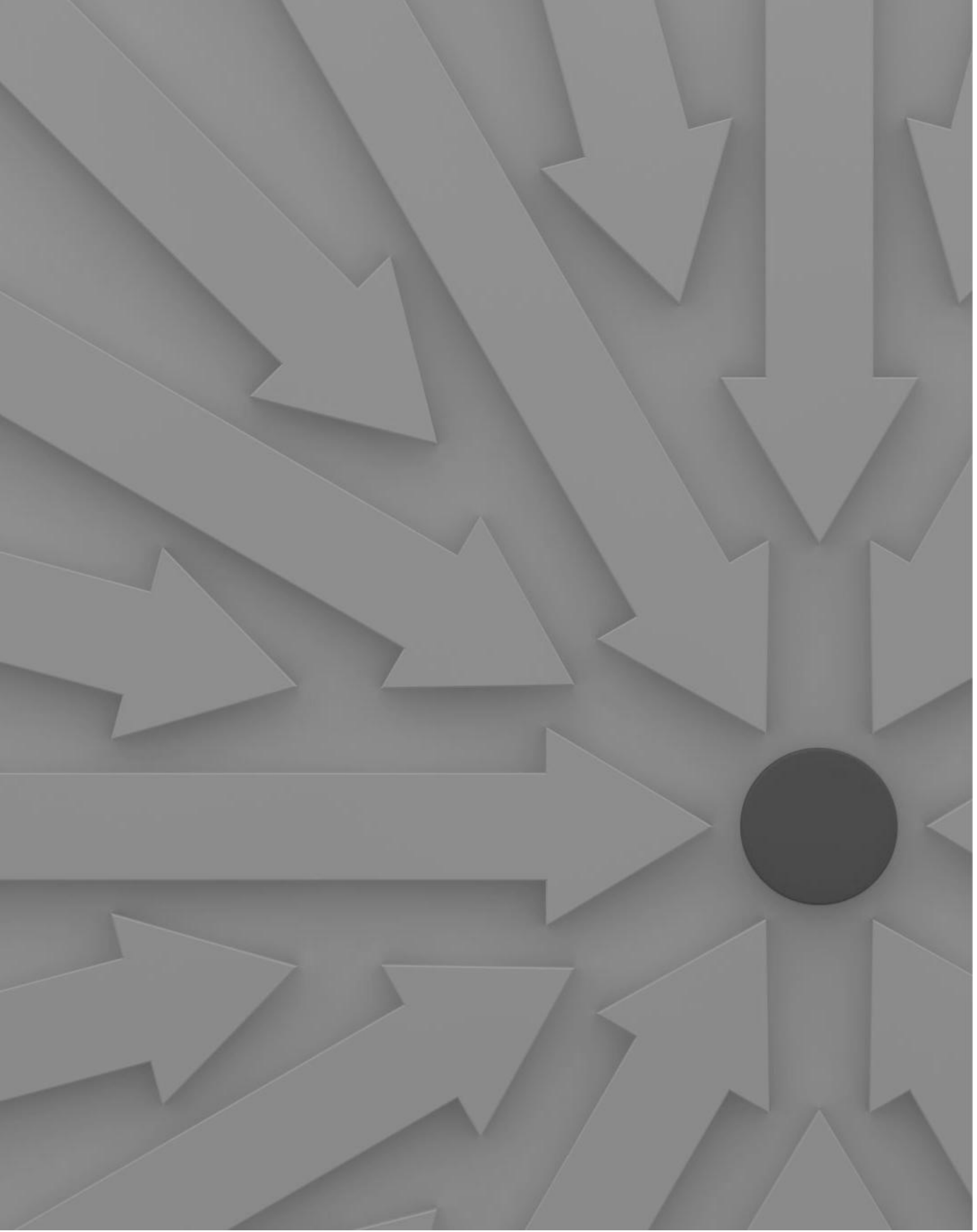
# DEHASHER: A HASH CRACKING PROJECT

MY PYTHON PROJECT FOR CY5001 FALL 2023

BY PRATIK DAS



# TABLE OF CONTENT

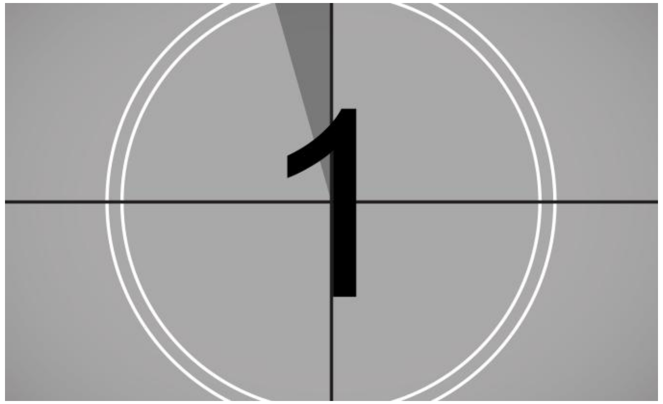


# OBJECTIVES

What does this project aim to do ?

Dehasher is a baseline project powered by Python to facilitate password recovery and assessment of password security by auditors and security professionals alike.

# USAGE




### Assess password strength

The hash can be tested against a list of known weak passwords and assorted credentials to observe if it returns its corresponding plaintext. This will validate the corresponding password's strength against credential brute forcing and guessing attacks.


### Recover plaintext password

With an effective and comprehensive wordlist, the provided hashes can be deciphered to their corresponding plaintext passwords.


### Multiple algorithms supported

The project leverages python's 'hashlib' library, which can assess password hash from the likes of MD5 to SHA256 algorithms.

# DEVELOPMENT PROCESS

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| RESEARCH | WRITING CODE | TESTING AND TROUBLESHOOTING | FINALIZING CODE |
| Searching about hashing algorithm and its implementation in python. | Breaking down the program into chunks of function and invoking them in appropriate sequence. | Running python code in consideration of arbitrary issues like wrong input, incorrect argument number and fixing the issues. | Documenting the code with comments to indicate functionality of each aspect. |

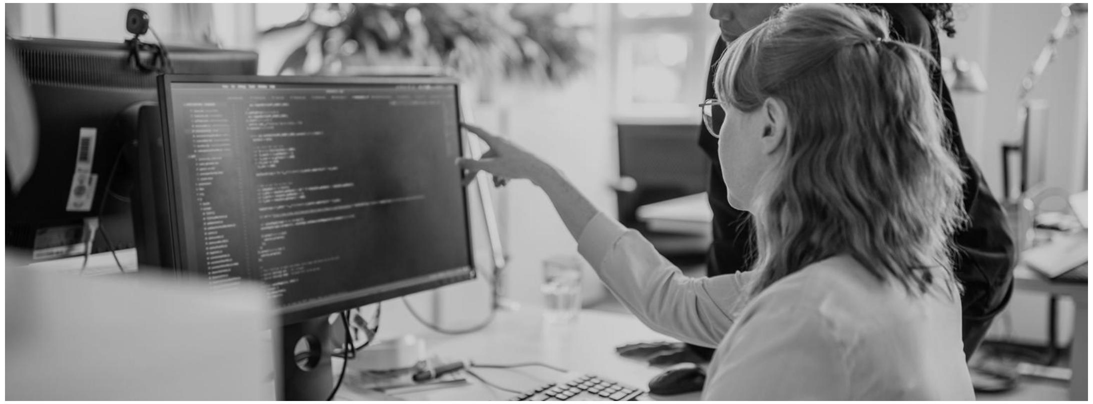
# DEVELOPMENT/IMPLEMENTATION ROADBLOCKS

- Rainbow table’s implementation required more time and in-depth research & understanding of concept.
- Switching from generic input to argument parsing with python module required testing and troubleshooting.
- Implementing parallelization and multi-threading resulted in complications at runtime.

# WHAT’S NEXT

PROSPECTS OF THIS PROJECT

# FEATURES TO BE IMPLEMENTED



## ‘RESUME’ FUNCTIONALITY

- The project, when passed the same hash and password list, should run from the same entry that it was interrupted at, during the previous running instance. This would facilitate saving the number of cracking attempts as the whole password list would not be used repetitively to crack the hash.

## LEVERAGING RAINBOW TABLES

- Rainbow tables is a pre-computed data structure mapping all used passwords to a value representing their hash values. This makes hash cracking a much faster process as the code will not have to compute for each plaintext for every comparison.

# THANK YOU

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A CY5001 PROJECT